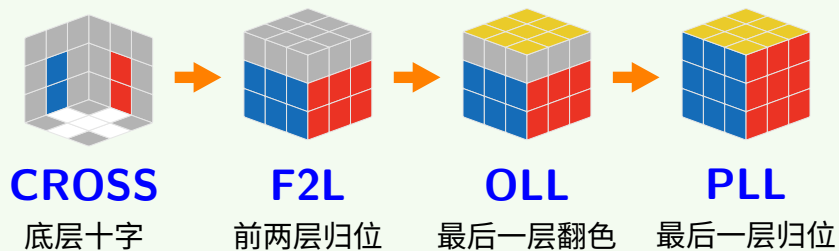




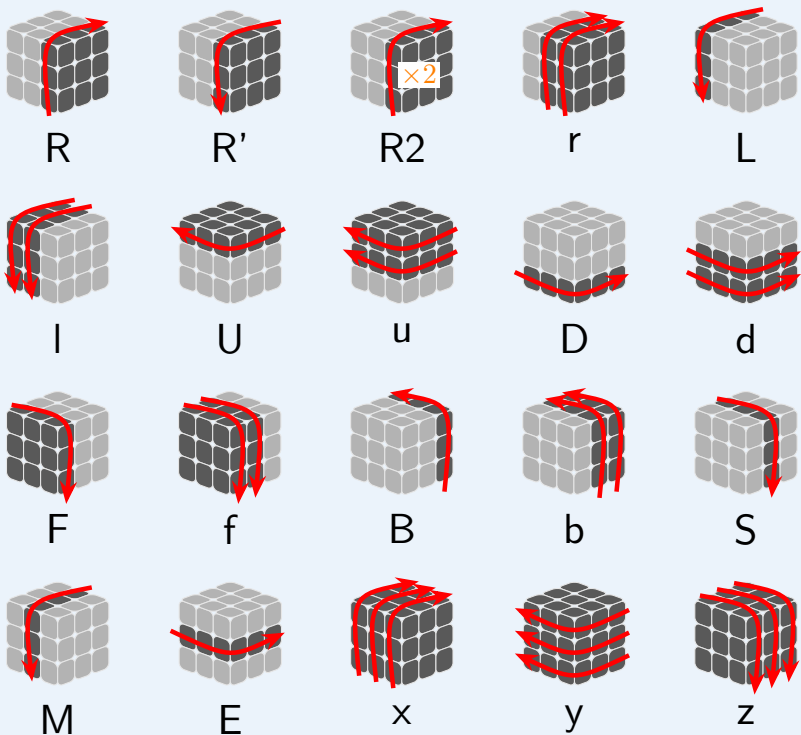
三阶魔方 CFOP 教程

3×3 CFOP TUTORIAL

CFOP 复原步骤



魔方转动公式符号及示意图



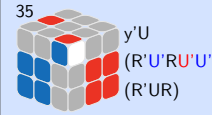
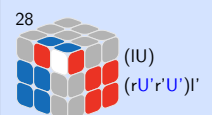
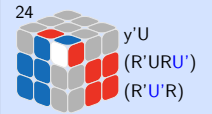
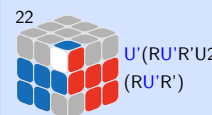
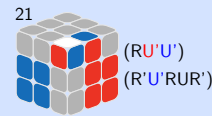
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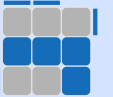
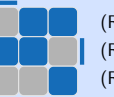
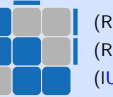
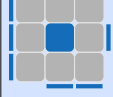
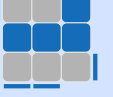
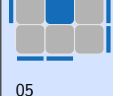
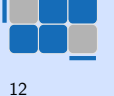
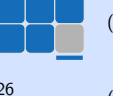
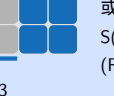
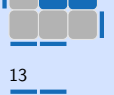
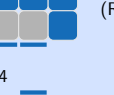
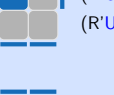
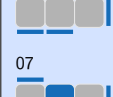
前两层归位

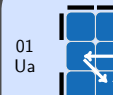
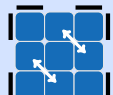
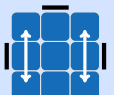
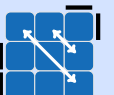
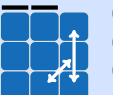
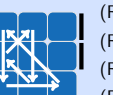
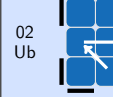
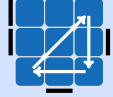
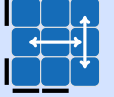
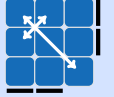
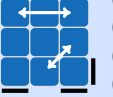
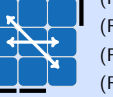
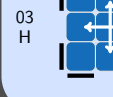
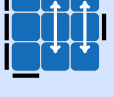
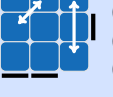
蓝色字母：用左食指拨动
红色字母：用右拇指拨动
绿色字母：用其它指拨动

小写字母：双层转
字母：顺时针转 90°
字母+：逆时针转 90°
字母+2：重复两次

温馨提示
*CROSS 复原方法较多，
本教程主要介绍后面 3 步
的复原公式



01  (R'U'U') (R'2FRF') U2(R'RF'F')	08  (r'U'RU') (R'U2r) 或者 y2l'U'LU' L'U2l	15  (r'U'r) (R'U'RU) (r'Ur)	22  (R'U'U') (R'2U')(R2U') (R'2U')(U'R)	29  (RUR'U') (RU'R'F'U'F) (RUR')	36  (R'U'RU') (R'URU) (lU'R'U)	43  (B'U') (R'URB) 或者 f'(L'U'LU)f	50  (r'U) (r2U'r'2U') (r2Ur')	57  (rUr') (URU'R')2 (rU'r') 或者 (fRUR'U'F') (RUR'U') (R'F'RF')
02  F(RUR'U')F' f(RUR'U')F'	09  (RUR'U') (R'F) (R2UR'U')F'	16  (rUr') (RUR'U') (r'U'r')	23  (R'2D') (RU'U'R'D) (RU'U'R)	30  f(RU) (R2U'R'U) (R2U'R')F'	37  F(RU'R'U') (RUR'F')	44  f(RUR'U')F'	51  f(RUR'U')2F'	OLL 最后一层翻色 OLL 和 PLL 是俯视图 OLL: 蓝块和小蓝条表示黄色块 PLL: 小黑条表示已归位的色块
03  f(RUR'U')f' U' F(RUR'U')F'	10  (RUR'U) (R'FRF') (RU'U'R')	17  (RUR'U) (R'FRF') U2(R'FRF')	24  (rUR'U') (r'FRF')	31  R'FRUR' U'F2UFR	38  (RUR'U) (RU'R'U') (R'FRF')	45  F(RUR'U')F'	52  (R'F'U'FU') (RUR'UR)	
04  f(RUR'U')F' f'U F(RUR'U')F'	11  (r'R2UR'U) (RU'U'R'U) M'	18  (rUR'U) (RU'U'r') (r'U'RU') (R'U2r)	25  F'(rUR'U') (r'FR)	32  (RU)(B'U') (R'URBR') 或者 S(RUR'U') (R'FRf')	39  (RUR'F'U'F) U(RU2R')	46  (R'U') (R'FRF') (UR)	53  (r'U2) (RUR'U') (RUR'Ur)	
05  (r'U2) (RUR'U)r 或者 y2l'U2L UL'UI	12  M'(R'U'RU') (R'U2RU')M	19  (r'RU) (RUR'U'r) (R'2FRF')	26  (RU'U') (R'U'RU'R') 或者 y'R'U'RU' RU2R'	33  (RUR'U') (R'FRF')	40  (R'F) (RUR'U') F'(UR)	47  b'(U'R'UR)2b 或者 F'(L'U'LU)2F	54  (rU'U') (R'U'RU) (R'U'RU'r')	
06  (rU'U') (R'U'RU'r') 或者 y2IU2L'U' LU'I'	13  f(RU) (R2U') (R'URU')f'	20  (rUR'U') M'2U (RU'R'U')M'	27  (RUR'U) (RU'U'R') 或者 y'R'U2R UR'UR	34  (RUR2U') (R'F) (RUR'U'F')	41  (RUR'U) (RU'U'R')F (RUR'U')F'	48  F(RUR'U')2F'	55  (rU'U'R'U') (r'R2UR'U') (rU'r')	
07  (rUR'U) (RU'U'r') 或者 y2IUL'U LU2l'	14  (R'F) (RUR'F'R) (FU'F')	21  (RU'U') (R'U'RUR'U') (RU'R')	28  (rUR'U') (r'RU) (RU'R')	35  (RU'U') (R'2FRF') (RU'U'R')	42  (R'U'RU') (R'U2R)F (RUR'U')F'	49  (RB') (R'2FR2B) (R'2F'R)	56  (RUR'U') M'(URU'r')	

01 Ua  M'2UMU2 M'UM'2 或者 (RU'R)(UR)2 (U'R'U'R'2)	04 Z  M'U (M'2U)2 M'U2 M'2U'	07 E  x'(RU'R'D) (RUR'D') (RUR'D) (RU'R'D')	10 V  R'URUR'f' (U'RU'2R') (U'RU'R')fR	13 Jb  (RUR'F') (RUR'U') (R'F) (R2U'R'U')	16 Gc  (R'2u'RU'R) (UR'u) (R2fr'f')	19 Gb  (R'd'F) (R2u) (R'U) (RU'Ru'R'2)
02 Ub  M'2U'MU2 M'U'M'2 或者 (R2U)(RU) (R'U')2(R'UR')	05 Aa  x'R2D2 (R'UR)D2 (R'UR')	08 T  (RUR'U') (R'F) (R2U'R'U') (RUR'F')	11 Y  F(RU'R'U') (RUR'F') (RUR'U') (R'FRF')	14 Rb  (R'U2) (RU'U') (R'FRUR'U') (R'F'R2U')	17 Gd  (RUR'y') (R2u'RU') (R'UR'uR2)	20 Nb  (R'URU') (R'F'U') (FRUR'F) (R'F'RU'R)
03 H  M'2UM'2 U2 M'2UM'2	06 Ab  x'(RU'R)D2 (R'UR)D2R'2	09 F  (RU'F') (RUR'U') (R'F) (R2U'R'U') (RUR'UR)	12 Ja  z(U'RD') (R2UR'U') (R2U) (DR')	15 Ra  (RU'R'U') (RURD) (R'U'RD') (R'U2R'U')	18 Ga  R2UR'UR' U'RU'R2 (U'D)R'UR (D'U)	21 Na  (RUR'U) (RUR'F') (RUR'U')(R'F) (R2U'R'U2) (RU'R')

PLL

最后一层归位